

# Van Doi Truong

vandoitruong1997@gmail.com ◇ Google Scholar ◇ LinkedIn  
(Updated Dec, 2025)

## EDUCATION

### Hanyang University

Master-PhD combined program in Mechanical Engineering

Ansan-si, Republic of Korea

03/2020 - 12/2025

### Ho Chi Minh University of Technology (HCMUT)

Bachelor of Science in Mechanical Engineering (8.68/10, Rank#1)

Ho Chi Minh city, Viet Nam

09/2015 - 09/2019

## RESEARCH EXPERIENCE

### Computational Hybrid Analysis of Materials Processing lab

- Currently developing a deep learning model for bead geometry prediction and monitoring in multi-pass, multi-layer welding
- Developed a machine vision system integrated with deep learning for hole expansion testing across various materials and thicknesses
- Developed a human-machine interaction system for automotive and medical imaging applications, enabling machine control through gestures and voice commands
- Developed a machine vision system for detecting defects in hardwood flooring and automotive parts during manufacturing

## PUBLICATION

### Journal articles

- **Van Doi Truong**, Jonghun Yoon\*, "Deep learning-based machine vision system for real-time edge fracture detection in the hole expansion ratio test.", 2026. (IF: 8, Q1), ref.
- **Van Doi Truong**, Yunfeng Wang, Chanhee Won, Jonghun Yoon\*, "A Deep Learning-Based Machine Vision System for Online Monitoring and Quality Evaluation During Multi-Layer Multi-Pass Welding.", 2025. (IF: 3.5, Q2), ref.
- Lixiao Zhou, **Van Doi Truong**, Jonghun Yoon\*, "Integrated ultra-wideband microwave system to measure composition ratio between fat and muscle in multi-species tissue types.", 2025. (IF: 3.5, Q2), ref.
- Yuhyeong Jeong, Yunfeng Wang, **Van Doi Truong**, Seungbeom Jang, Jangwook Lee, Jonghun Yoon\*, "Ensemble machine learning model for prediction of inner welding defects in orbital GTAW (Gas Tungsten Arc Welding) process with monitored by multi-sensor system.", 2024. (IF: 3.5, Q1), ref.
- **Van Doi Truong**, Hyun-Kyo Lim, Seongje Kim, Than Trong Khanh Dat, Jonghun Yoon\*, "Contagious infection-free medical interaction system with machine vision controlled by remote hand gesture during an operation," *Computational and Structural Biotechnology Journal*, 2024. (IF: 4.5, Q1), ref.
- Seongje Kim, **Van Doi Truong**, Kwang-Hee Lee, Jonghun Yoon\*, "Revolutionizing Robotic Depalletizing: AI-Enhanced Parcel Detecting with Adaptive 3D Machine Vision and RGB-D Imaging for Automated Unloading," *Sensors*, 2024. (IF: 3.4, Q2), ref.
- Yunfeng Wang, Wonjoo Lee, Seungbem Jang, **Van Doi Truong**, Yuhyeong Jeong, Chanhee Won, Jangwook Lee, Jonghun Yoon\*, "Prediction of internal welding penetration based on IR thermal image supported by machine vision and ANN-model during automatic robot welding process," *Journal of Advanced Joining Processes*, 2024. (IF: 3.8, Q1), ref.
- **Van Doi Truong**, Jiaping Xia, YuHyeong Jeong, Jonghun Yoon\*, "An automatic machine vision-based algorithm for inspection of hardwood flooring defects during manufacturing," *Engineering Applications of Artificial Intelligence*, 2023. (IF: 8, Q1), ref.
- **Van Doi Truong**, Yong-Taek Hyun, Jong Woo Won, Wonjoo Lee, Jonghun Yoon, "Numerical simulation of the effect of scanning strategies on the aluminum evaporation of titanium alloy in the electron beam cold hearth melting process," *Materials*, 2022. (IF: 3.1, Q2), ref.

### Papers in Preparation

- **Van Doi Truong**, Jonghun Yoon\*, "Deep learning prediction of weld bead geometry during multi-pass multi-layer welding process."

## CONFERENCE PRESENTATIONS

---

- A vision-based system for weld bead profile extraction during multi-layer multi-pass welding process, Oral presentation delivered at the *The 76th Annual Assembly and International Conference on Welding and Joining (IIW 2023)*, Singapore, 2023.
- Medical interaction system with machine vision and remote gestures for infection free, Poster presentation delivered at the *KSPE 2023 Autumn Conference*, Korea, 2023.
- Automatic welding defect inspection during orbital welding using multi-sensor and deep learning approach, Oral presentation delivered at the *17th International Conference on Advanced Computational Engineering and Experimenting (ACEX2024)*, Spain, 2024.

## AWARDS

---

- Silver medal in Kinematics and Dynamics of Machines, National Mechanics Olympiad, Viet Nam, 04/2016
- Bronze medal in Applied Informatics in Machine Design, National Mechanics Olympiad, Viet Nam, 04/2017
- Science and Engineering Revitalization Scholarship, Hanyang University, 03/2020 - 03/2023
- BrainKorea21 Four (BK21FOUR) Scholarship, Hanyang University, 03/2022 03/2023
- 2nd BK Excellent Academic Activity Award, Hanyang University, 03/2024